Chapter 10

How to use Grid Layout for Responsive Web Design

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Objectives

Applied

1. Use Grid Layout to lay out web pages or components within web pages, including the layouts for mobile devices.

Knowledge

1. In general terms, describe the use of Grid Layout.
2. Describe these terms as they apply to Grid Layout: grid, grid track, grid line, grid cell, grid area, and grid item.
3. In general terms, describe these ways to define the grid areas for the elements of a page: numbered lines, template areas, and the 12-column grid concept.

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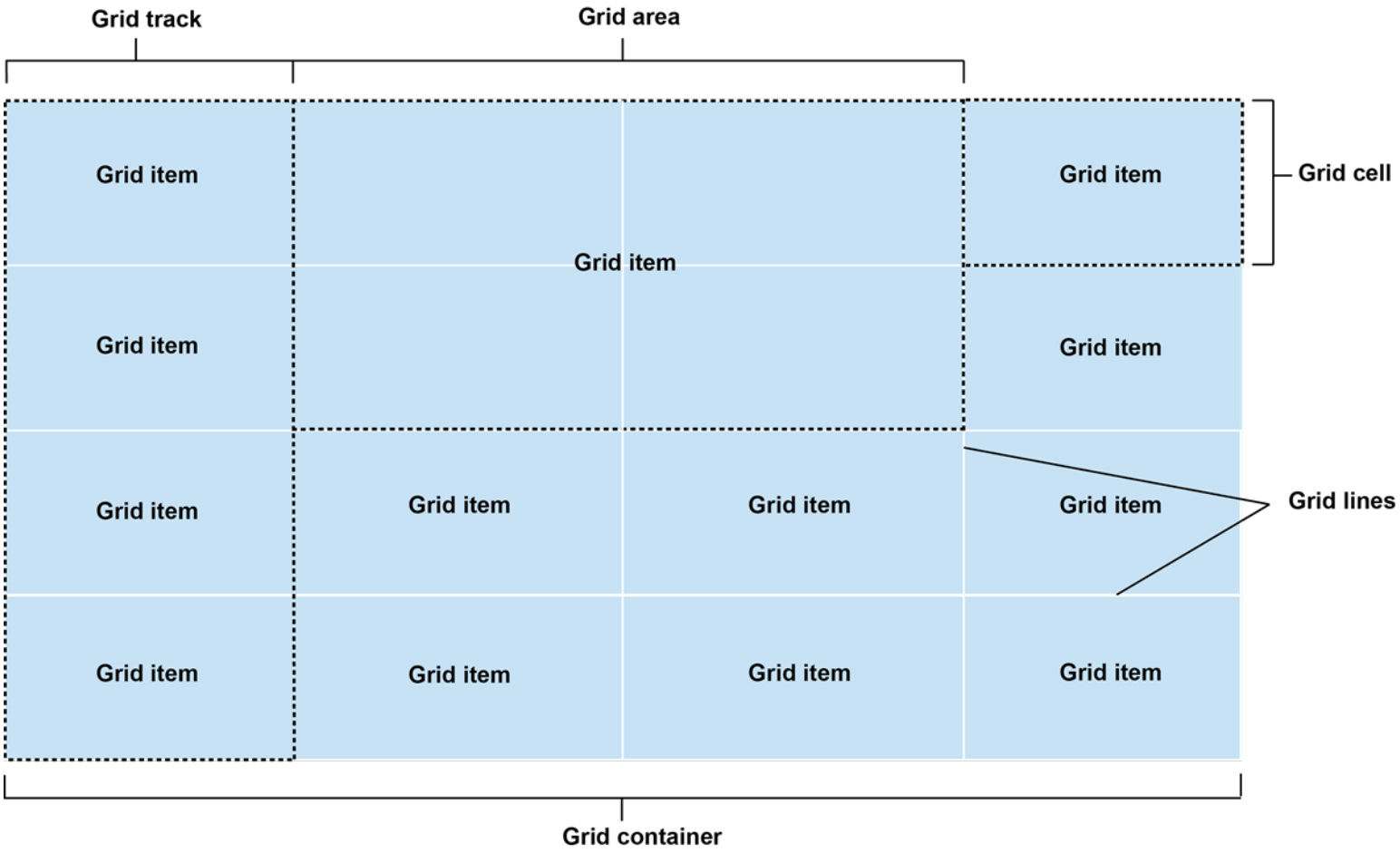
The components of a grid layout



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The HTML and CSS for a 2 row by 3 column grid

The HTML

```
<main>
    <div>Div 1</div> <div>Div 2</div> <div>Div 3</div>
    <div>Div 4</div> <div>Div 5</div> <div>Div 6</div>
</main>
```

The CSS for laying out the grid

```
div { background-color: #C8DFEE; }
main {
    display: grid;
    grid-template: 100px 100px / 150px 150px 150px;
    grid-gap: 15px 20px; }
```


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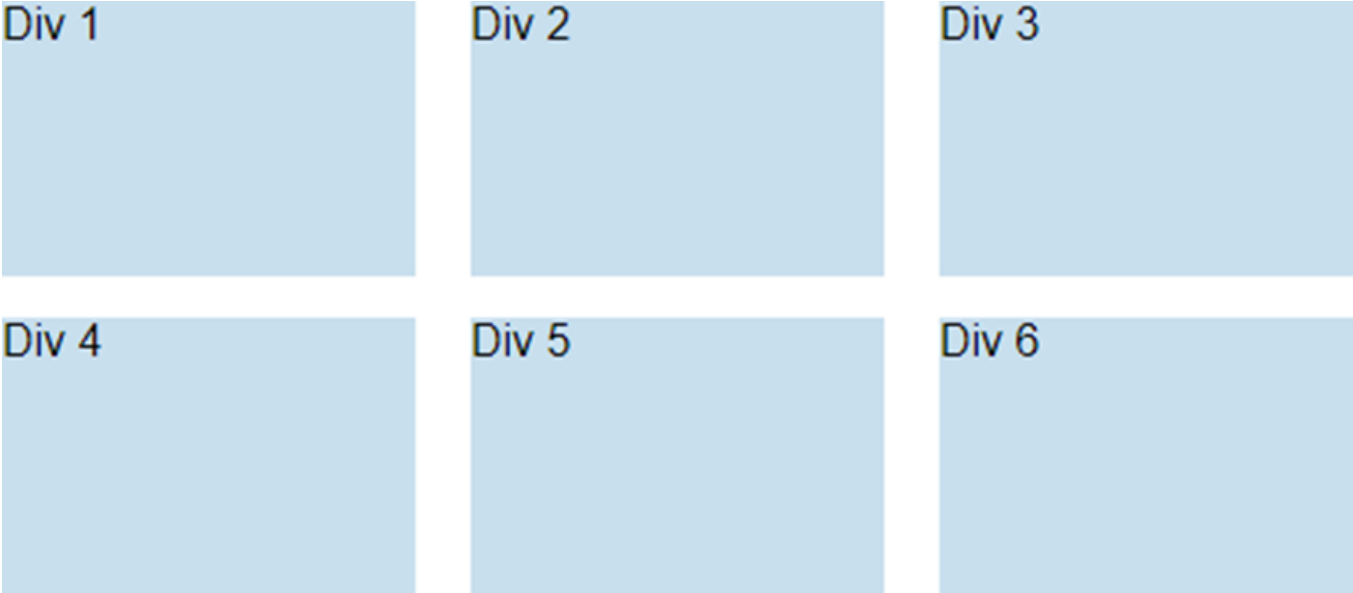
The layout for the 2x3 grid



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CSS that uses the repeat() function

```
main {
    display: grid;
    grid-template: repeat(2, 50px) / repeat(3, 1fr);
    grid-gap: 20px; }
```

The resulting layout

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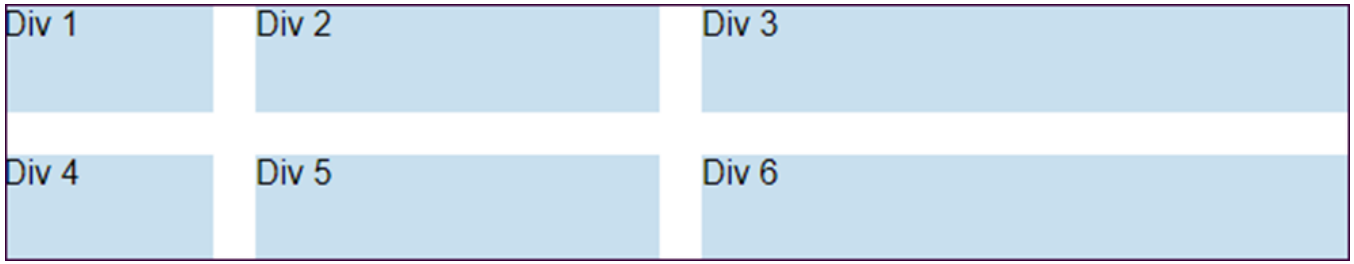
CSS that uses mixed units

```
main {
    display: grid;
    grid-template: repeat(2, 50px) / 100px 30% 1fr;
    grid-gap: 20px; }
```

The resulting layout



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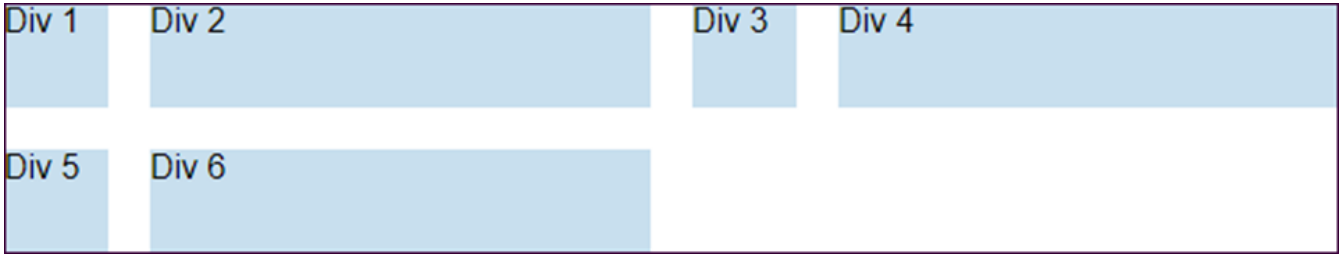
CSS that repeats two columns

```
main {
    display: grid;
    grid-template:
        repeat(2, 50px) / repeat(2, 50px 1fr);
    grid-gap: 20px; }
```

The resulting layout



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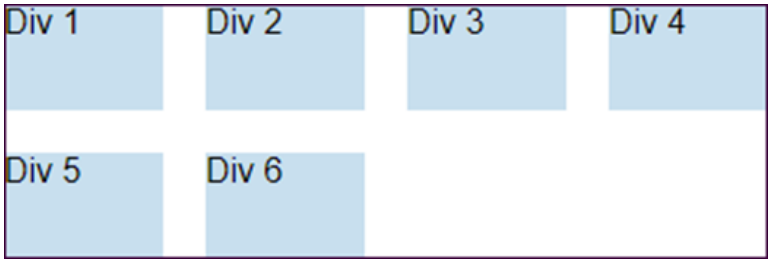
CSS that uses auto-fit and the minmax() function with the repeat() function

```
main {
  display: grid;
  grid-template:
    repeat(2, 50px) / repeat(auto-fit, minmax(75px, 1fr));
   grid-gap: 20px; }
```

The layout when the container is 360 pixels wide



The layout when the container is 720 pixels wide

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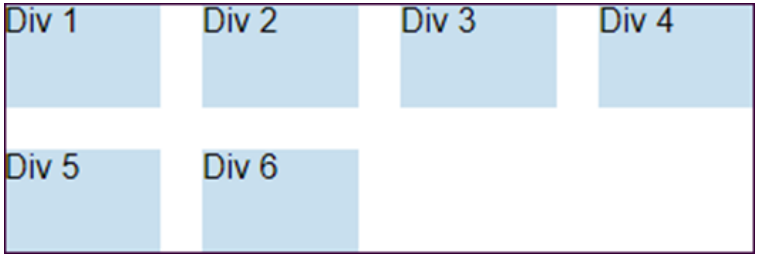
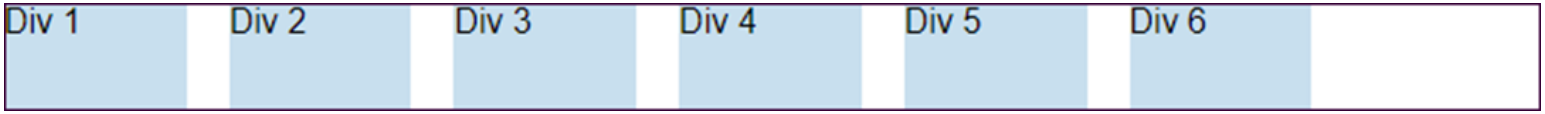
CSS that uses auto-fill and the minmax() function

```
main {
  display: grid;
  grid-template:
    repeat(2, 50px) / repeat(auto-fill, minmax(75px, 1fr));
  grid-gap: 20px; }
```

The layout when the container is 360 pixels wide



The layout when the container is 720 pixels wide



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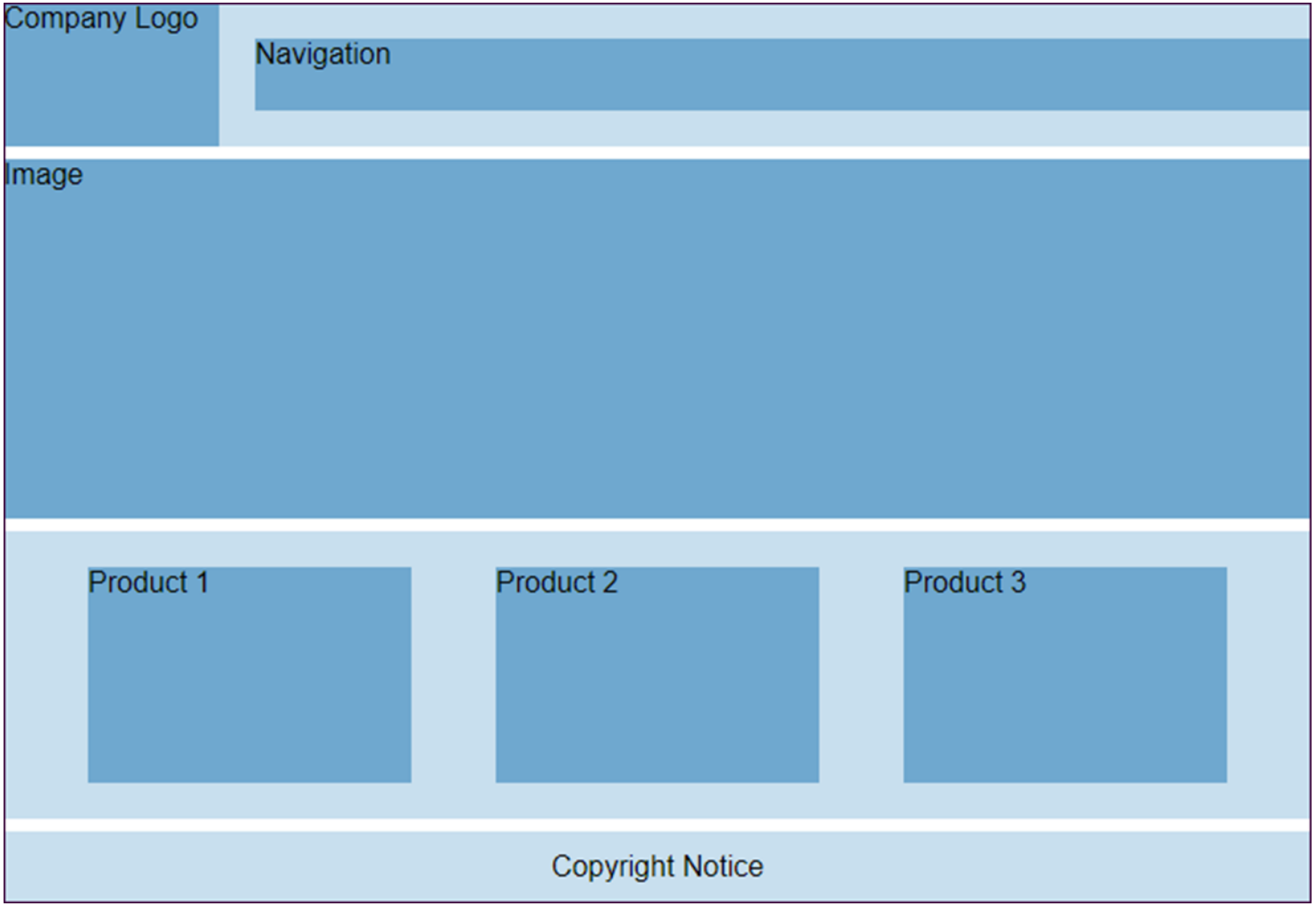
A page layout that uses alignment



The HTML for the body of the page

```
<body>
    <nav>
        <div>Company Logo</div>
        <div>Navigation</div>
    </nav>
    <section>
        <div>Image</div>
    </section>
    <main>
        <div>Product 1</div>
        <div>Product 2</div>
        <div>Product 3</div>
    </main>
    <footer><p>Copyright Notice</p></footer>
</body>
```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Some of the CSS for the page

```
nav, section, main, footer {
    display: grid;
    width: 100%; }
nav {
    grid-template: 80px / 120px 1fr;
    grid-gap: 20px; }
nav div:nth-of-type(2) {
    height: 40px;
    align-self: center; }
section { grid-template-rows: 200px; }
main {
    height: 160px;
    grid-template: 120px / repeat(3, 180px);
    align-content: center;
    justify-content: space-evenly; }
footer {
    height: 40px;
    align-items: center;
    justify-items: center; }
```


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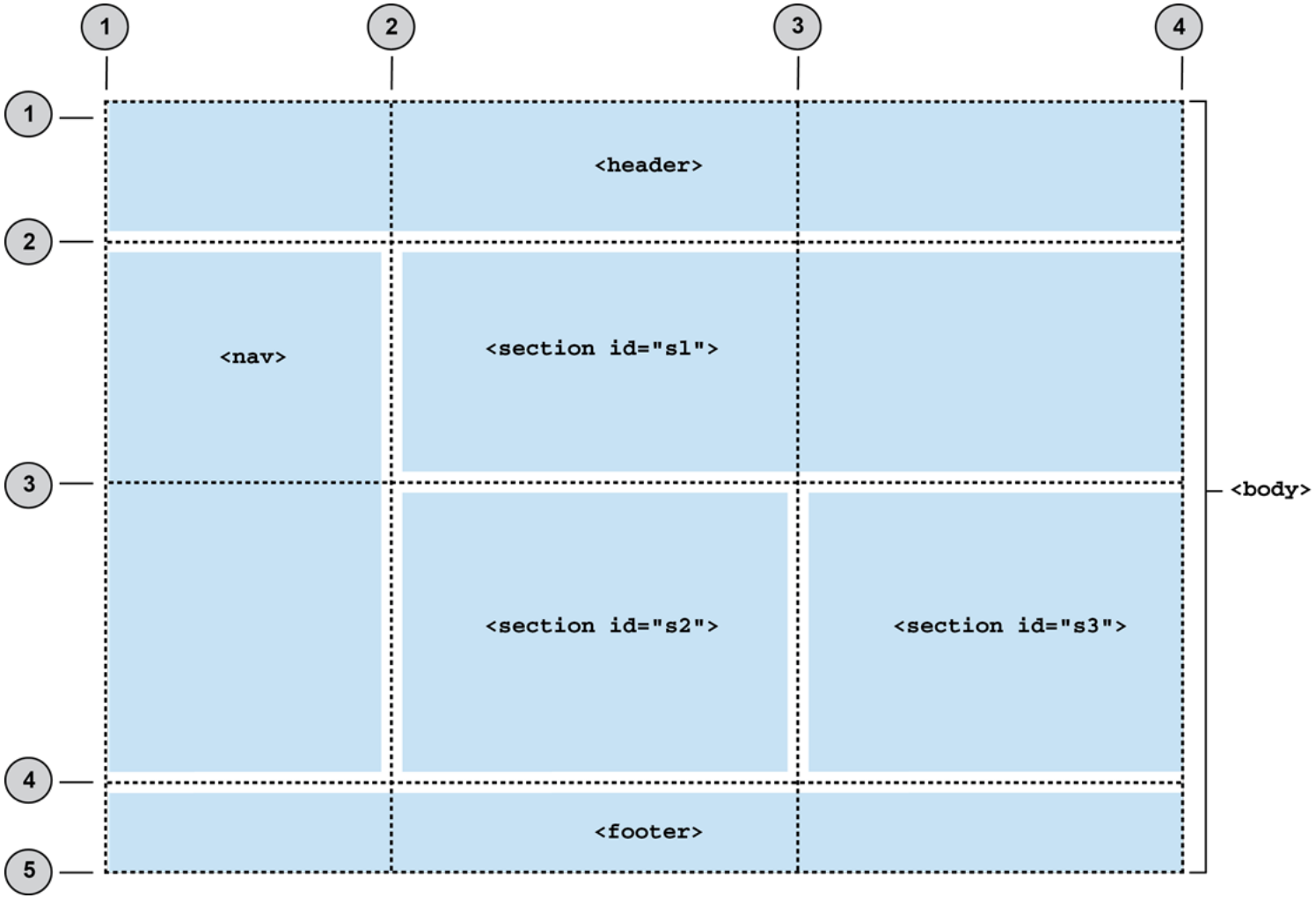
Numbered grid lines for a grid container



Numbered lines that define the grid areas

```
body {
    margin: 0;
    height: 600px;
    width: 100%;
    display: grid;
    grid-template-columns: 200px 1fr 1fr;
    grid-template-rows: 80px 1fr 1fr 60px;
    grid-gap: 16px; }
header { grid-row: 1 / 2; grid-column: 1 / 4; }
nav    { grid-row: 2 / 4; grid-column: 1 / 2; }
#s1    { grid-row: 2 / 3; grid-column: 2 / 4; }
#s2    { grid-row: 3 / 4; grid-column: 2 / 3; }
#s3    { grid-row: 3 / 4; grid-column: 3 / 4; }
footer { grid-row: 4 / 5; grid-column: 1 / 4; }
```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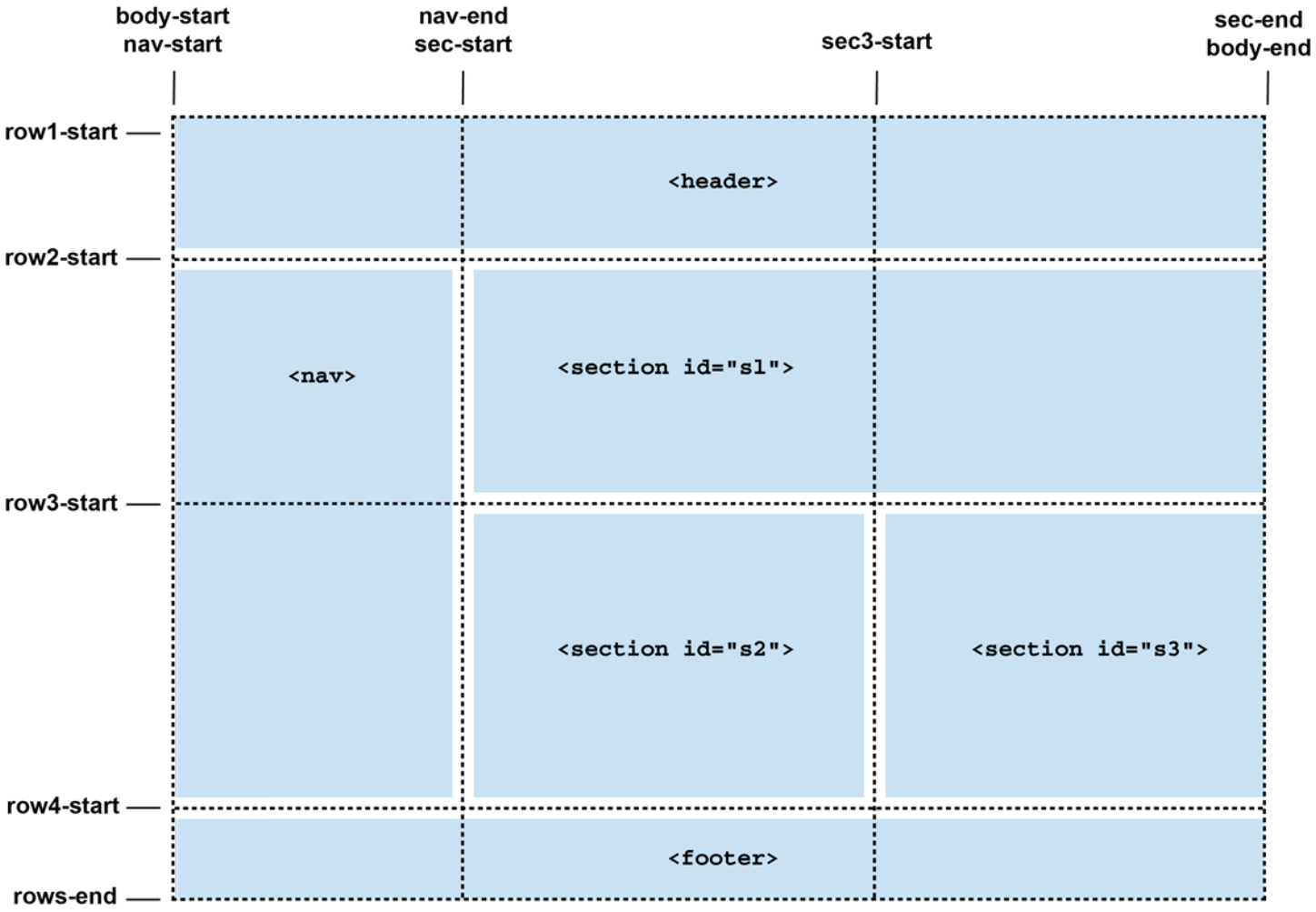
Named lines for a grid container



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Named lines that define the grid areas

```
body {
    display: grid;
    grid-template-columns: [body-start nav-start] 200px
        [nav-end sec-start] 1fr
        [sec3-start] 1fr [sec-end body-end];
    grid-template-rows: [row1-start] 80px [row2-start] 1fr
        [row3-start] 1fr [row4-start] 60px [rows-end];
    grid-gap: 16px; }
header { grid-row: row1-start/row2-start;
         grid-column: body-start/body-end; }
nav    { grid-row: row2-start/row4-start;
         grid-column: nav-start/nav-end; }
#s1    { grid-row: row2-start/row3-start;
         grid-column: sec-start/sec-end; }
#s2    { grid-row: row3-start/row4-start;
         grid-column: sec-start/sec3-start; }
#s3    { grid-row: row3-start/row4-start;
         grid-column: column3-start/content-end; }
footer { grid-row: row4-start/rows-end;
         grid-column: body-start/body-end; }
```


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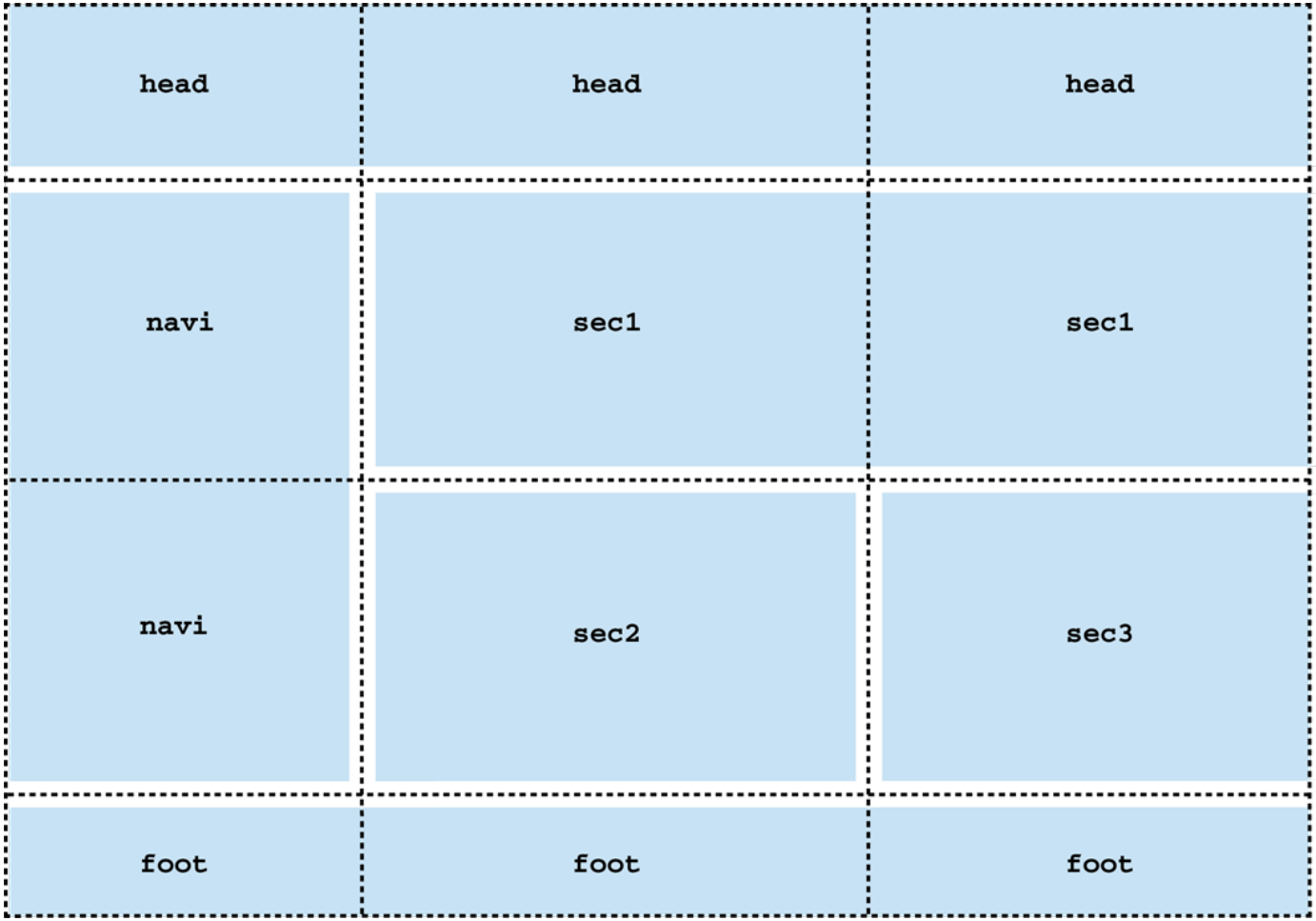
The template names for the cells within a grid container



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Using template names to define the grid areas

```
body {
    display: grid;
    grid-template-columns: 200px 1fr 1fr;
    grid-template-rows: 80px 1fr 1fr 60px;
    grid-gap: 16px;
    grid-template-areas:
        "head head head"
        "navi sec1 sec1"
        "navi sec2 sec3"
        "foot foot foot"; }
header { grid-area: head; }
nav    { grid-area: navi; }
#s1    { grid-area: sec1; }
#s2    { grid-area: sec2; }
#s3    { grid-area: sec3; }
footer { grid-area: foot; }
```


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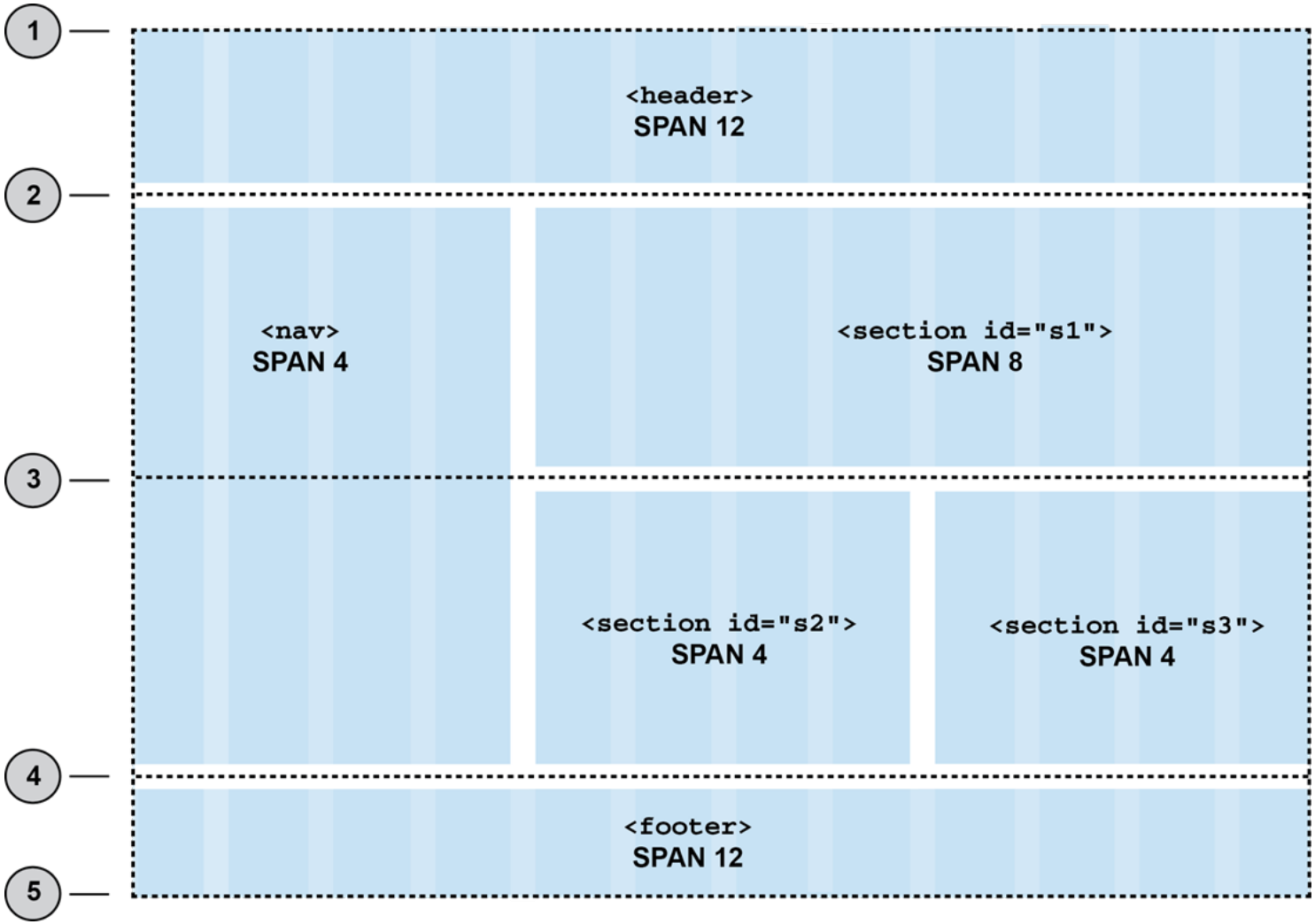
The 12-column grid for a grid container



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Using a 12-column grid to define the grid areas

```
body {
    display: grid;
    grid-template-columns: repeat(12, 1fr);
    grid-template-rows: 80px 1fr 1fr 60px;
    grid-gap: 16px; }
header { grid-row: 1 / 2; grid-column: 1 / span 12; }
nav    { grid-row: 2 / 4; grid-column: 1 / span 4; }
#s1    { grid-row: 2 / 3; grid-column: 5 / span 8; }
#s2    { grid-row: 3 / 4; grid-column: 5 / span 4; }
#s3    { grid-row: 3 / 4; grid-column: 9 / span 4; }
footer { grid-row: 4 / 5; grid-column: 1 / span 12; }
```


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A speaker page in desktop layout

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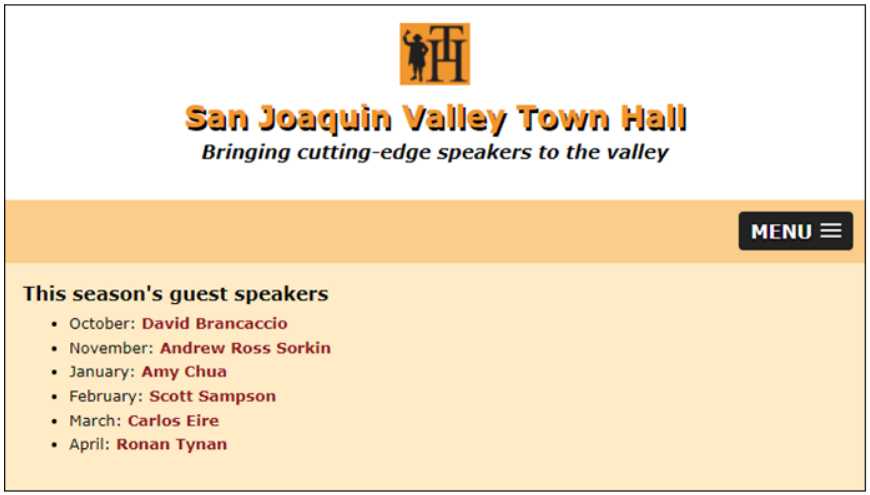
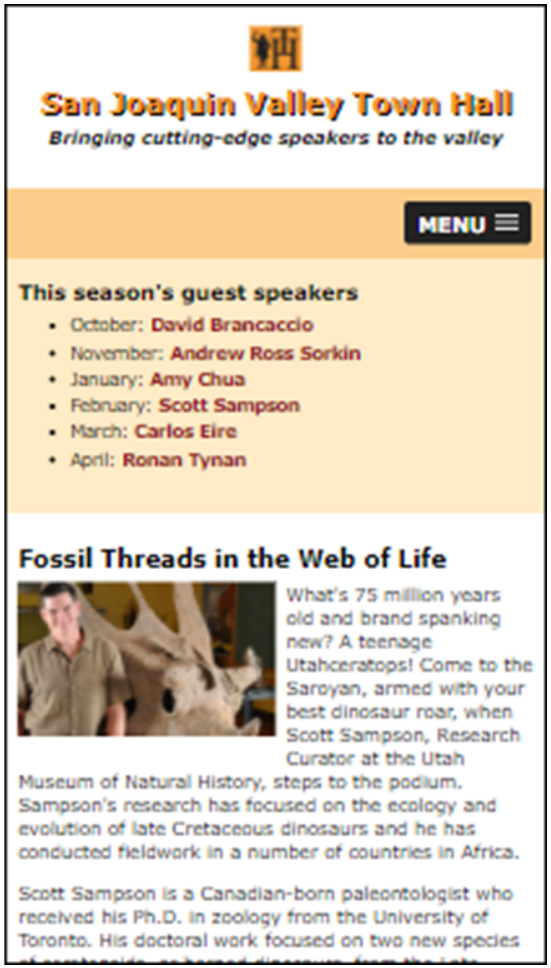
A speaker page in mobile layout





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The HTML for the structural elements

```
<body>
    <header>...</header>
    <nav id="mobile_menu"></nav>
    <nav id="nav_menu">
        <ul>
            <li><a href="index.html">Home</a></li>
            ...
        </ul>
    </nav>
    <main>
        <article>...</article>
        <aside>
            <h2>This season's guest speakers</h2>
            <nav>
                <ul>   </ul>
            </nav>
        </aside>
    </main>
    <footer>...</footer>
</body>
```


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CSS that uses template areas

```
body {
    ...
    display: grid;
    grid-template: repeat(4, auto) / 1fr;
    grid-gap: 1.5em;
    grid-template-areas:
        "head"
        "navi"
        "main"
        "foot"; }
header {
    grid-area: head; }
#nav_menu {
    grid-area: navi; }
#nav_menu ul {
    ...
    display: grid;
    grid-template: auto / repeat(5, 1fr);
    align-content: center; }
```


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CSS that uses template areas (continued)

```
main {
    margin: 0 2%;
    grid-area: main;
    display: grid;
    grid-template: auto / 1fr 40%;
    grid-gap: 2%;
    grid-template-areas:
        "arti side"; }
article {
    grid-area: arti; }
aside {
    max-height: 200px;
    min-height: 180px;
    padding: 5%;
    background-color: #ffebc6;
    grid-area: side; }
footer { grid-area: foot; }
```


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The media query for smaller screens that uses template areas

```
@media only screen and (max-width: 796px) {
    body {
        ...
        grid-gap: 0; /* no space between menu and main */
    }
    header {
        padding-bottom: 1.5em;  /* because no row gap */
        display: grid;          /* make header a grid */
        grid-template: repeat(3, auto) / 1fr;
        grid-gap: .3em;
        justify-items: center;  /* center horizontally */
    }
    #mobile_menu {
        display: block;
        grid-area: navi;
    }
```


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The media query for smaller screens that uses template areas (continued)

```
main {
    margin: 0;
    grid-template: auto auto / 1fr;
    grid-gap: 2%;
    grid-template-areas:
        "side"
        "arti";
}
```


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The CSS for a 12-column grid

```
body {
    ...
    display: grid;
    grid-template: repeat(4, auto) / repeat(12, 1fr);
    grid-gap: 1.5em;
}
header {
    grid-row: 1 / 2;
    grid-column: 1 / span 12;
}
#nav_menu {
    grid-row: 2 / 3;
    grid-column: 1 / span 12;
}
```


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The CSS for a 12-column grid (continued)

```
article {
    margin-left: 3.4%;
    grid-row: 3 / 4;
    grid-column: 1 / span 7;
}
aside {
    ...
    margin-right: 5%;
    grid-row: 3 / 4;
    grid-column: 8 / span 5;
}
footer {
    ...
    grid-row: 4 / 5;
    grid-column: 1 / span 12;
}
```


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The media query for smaller screens that uses a 12-column grid

```
@media only screen and (max-width: 796px) {
    body {
        ...
        grid-template: repeat(5, auto) / 1fr;
        grid-gap: 0;
    }
    header {
        grid-column: 1 / 2;
        ...
    }
    #mobile_menu {
        display: block;
        grid-row: 2 / 3;
        grid-column 1 / 2;
    }
```


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The media query for smaller screens that uses a 12-column grid (continued)

```
        article {
            ...
            grid-row: 4 / 5;
            grid-column: 1 / 2;
        }
        aside {
            margin-right: 0;
            padding: 1em 2%;
            grid-row: 3 / 4;
            grid-column: 1 / 2;
        }
        footer {
            grid-row: 5 / 6;
            grid-column: 1 / 2;
        {
    }
```


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The headline and gallery layout

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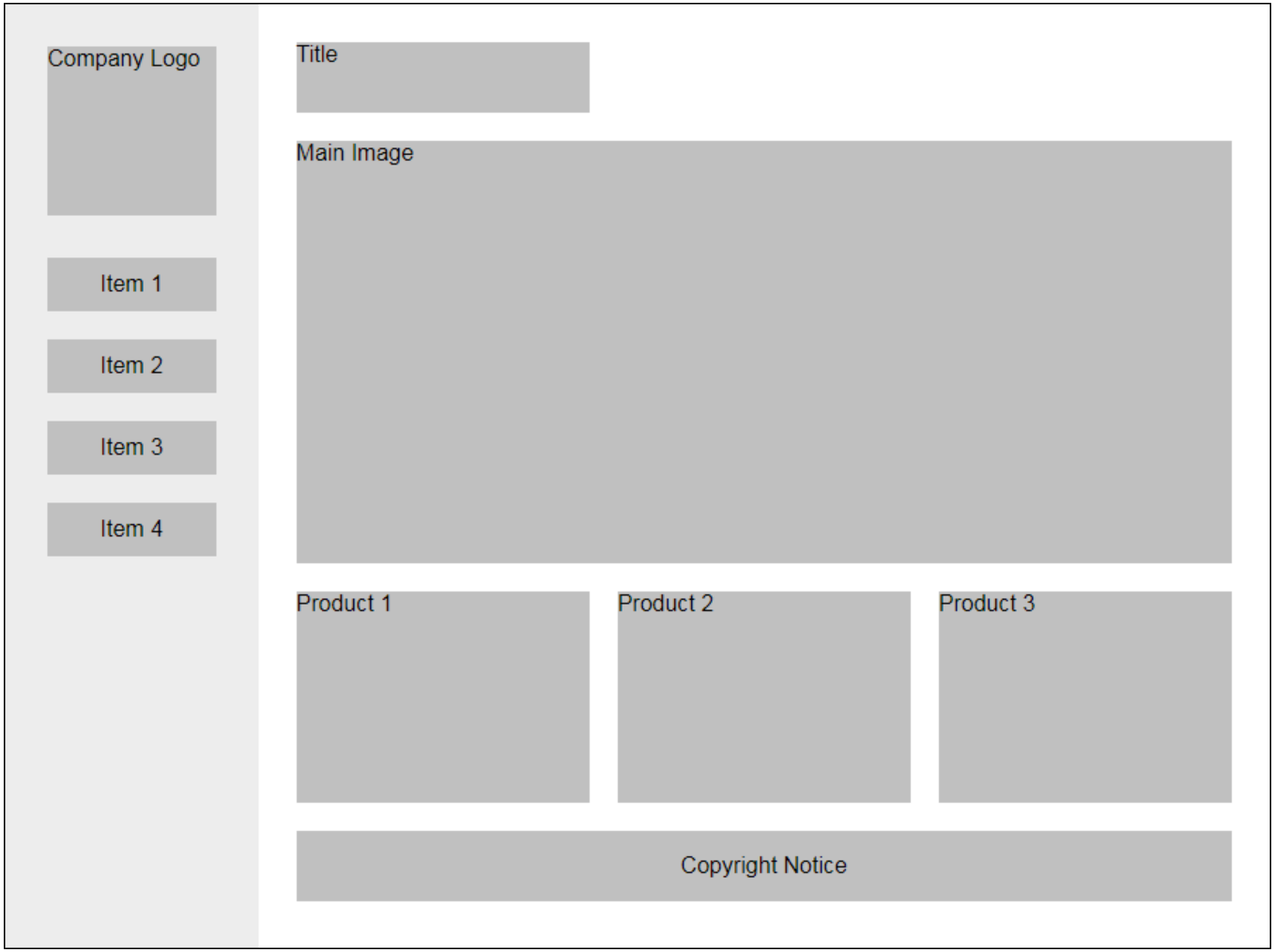
The fixed sidebar layout



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The advanced grid layout

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Short 10-1 Use a 12-column grid for the Town Hall home page

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